Title: Coursework Assignment

Course: ST3189 Machine Learning

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Essay Word Count: 2,537

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1. Background

In adverse social, economic, cultural and family conditions, some individuals are coerced to commit crimes that severely impacts the safety, wellbeing and stability of their community. These crimes not only affect direct victims but influences the fabric of society, such as social harmony and trust in law enforcement and government institutions.

Due to the vast geographic diversity and socioeconomic disparities, the United States (US) has exhibited a wide range of crimes, from violent offenses such as assault and murder to property crimes like burglary and theft. Crimes have only been made more accessibly with the accessibility and advancements in technology. With the diverse nature of crimes and varying intensities across different regions, it is crucial to understand crime patterns to develop effective strategies to mitigate its effects.

In recent years, government agencies in the US adopted the National Incident-Based Reporting System (NIBRS) for reporting crimes and arrests. This system opens opportunities to analyze and gain invaluable insights about crime characteristics and trends. Moreover, the National Crime Victimization Survey (NCVS) by the US Census Bureau in 2022 offers additional in-depth information into the nature and frequency of crime, beyond what is reported to law enforcement. However, despite the availability of data and information, crime remains a pressing issue, highlighting the need for interventions and innovative solutions.

2. Research Questions

Utilizing crime data collected by the Los Angeles Police Department (LAPD), this research will analyse incidents of crime in LA from 2020 to 2024 (Achary, 2021). Using demographic characteristics provided in the datasets, this research hopes to discover patterns and trends about the victims of crimes from 2020 to 2024. Additionally, descriptive data about the crimes committed and weapons used can help to uncover potential relationships and correlation between the variables. Furthermore, this research hopes to answer these questions using machine learning tools:

1) What clusters can be observed from crimes committed in LA from 2020 to 2024?
2) Can the demographic traits of victims of crimes in LA in 2022 be predicted?
3) Can the latitude and longitude of crime locations in LA in 2022 be predicted?

The results will be supplemented by and compared against reports and studies of victimisation statistics in 2022 by NIBRS and NCVS to draw meaningful conclusions about crime in the US. Using the results, this research aims to develop strategies to directly reduce or mitigate targeted crime.

3. Exploratory Data Analysis (EDA)

3.1. EDA on Crimes from 2020 to 2024

Using data from 2020 to 2024, this research will focus on the biological sex and ethnic descents of victims as well as the nature of crimes to discover key trends of these features over the years. Values from the original dataset are modified and re-categorised to form a clearer analysis of the nature of crimes and their victimisations. As such, ethnic descents have mainly been categorised to Hispanic, White, Black, Asian, Pacific Islander, Native American, Others, and Unknown (*U.S. Census Bureau quickfacts: United States*). Next, weapons used have been categorized to no weapons, bodily force, non-blunt weapons, striking weapons,

sharp weapons, and firearms, where non-blunt weapons include dangerous weapons such as explosives. Lastly, crimes have been categorized to violent, statutory, sexual, financial, and theft, where theft crimes refers to theft on individuals or property such as robbery or shoplifting (*Types of criminal offenses*, 2023). Using these variables, an analysis can be made on the frequency of crimes committed from 2020 to 2024.

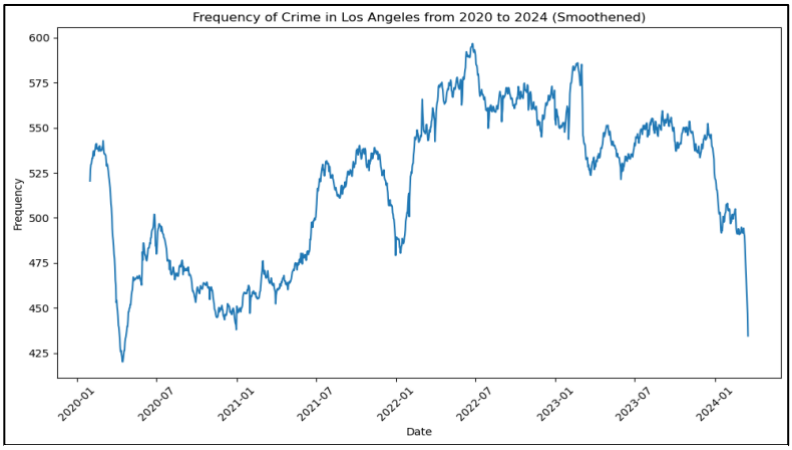


Figure 1. Frequency of Crime in Los Angeles from 2020 to 2024

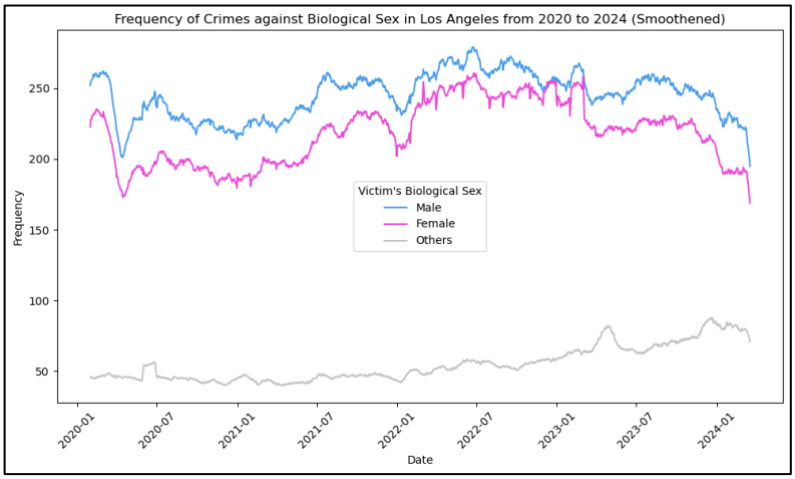


Figure 2. Frequency of Crimes against Biological Sex in Los Angeles from 2020 to 2024

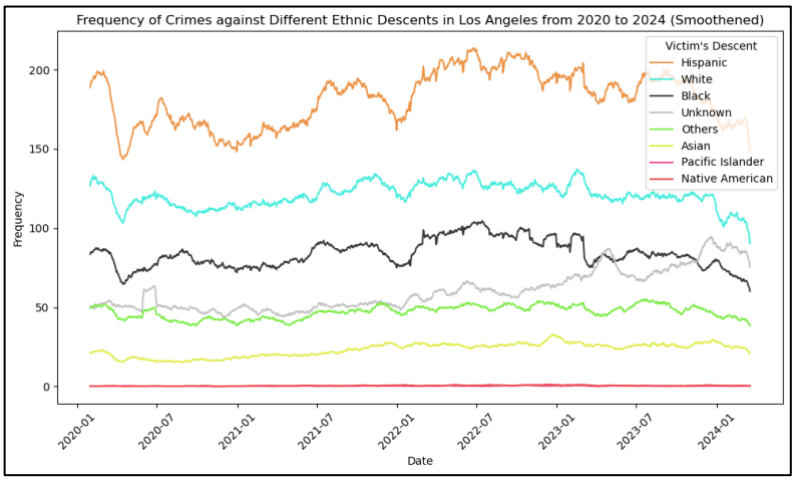


Figure 3. Frequency of Crimes by Ethnicity in Los Angeles from 2020 to 2024

Over the years, the frequency of crime increased from 2020 to 2022 but shows a downward trend from 2022 to 2024 (Figure 1). Similar trends can be observed about crimes committed against males, females, and majority of the ethnic descents (Figure 2 & 3). However, crimes committed against victims with a biological sex of 'Others' and an ethnic descent of 'Unknown' show an increasing frequency throughout the years. Through a drill down on the victims with a biological sex of 'Others' and an ethnic descent of 'Unknown', it is observed that more than 90% involved theft on property. Thus, we can conclude that there is an increasing number of property crimes.

Throughout the years, Males suffer from a greater number of crimes than Females. In terms of ethnic descent, Hispanics suffer from the greatest number of crimes, followed by White and Black victims.

Meanwhile, no distinct trend can be observed about the nature of crimes except crimes involving no weapons and theft crimes. Reflecting the trend of crimes in general, the frequency of crimes without weapons have increased over the years but decreased from 2023 onwards. Expectedly, the increasing frequency of theft

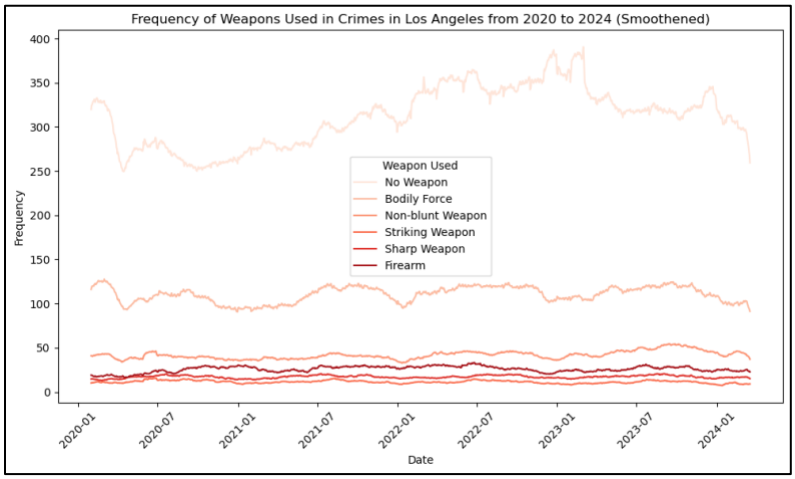


Figure 4. Frequency of Crimes involving Weapons in Los Angeles from 2020 to 2024

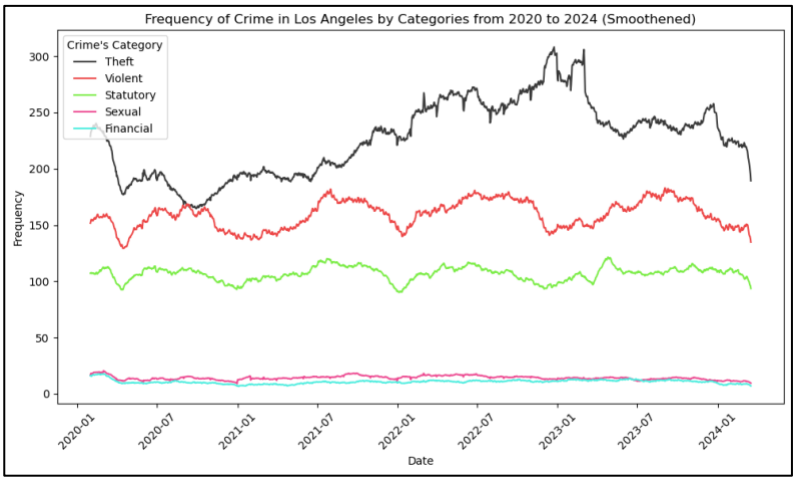


Figure 5. Frequency of Crimes by Categories in Los Angeles from 2020 to 2024

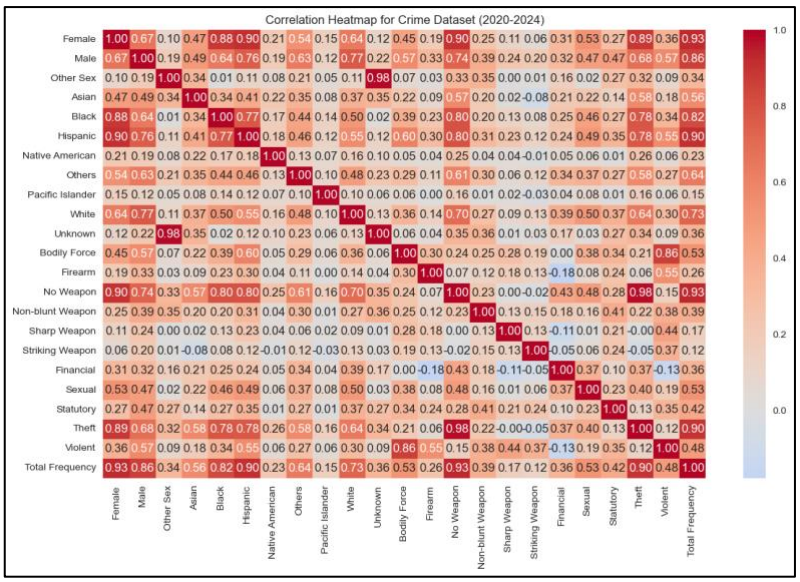


Figure 6. Correlation Heatmap

crimes throughout 2022 support the trend of crimes on 'Others' sex and 'Unknown' descent.

While analysis of the demographics and nature of crimes help to gain insights of the crimes committed from 2020 to 2024, supplementary analysis is needed to form conclusions about these insights. Using a correlation heatmap, we can draw meaningful conclusions about the relationship between different variables and develop innovative solutions to the rampant crime rate in the US. Figure 6 shows a strong positive correlation between these following variables: Female and Black (0.88), Female and Hispanic (0.90), Female and No Weapon (0.90), Female and Theft (0.89), No Weapon and Black (0.80), No Weapon and Hispanic (0.80), Violent and Bodily Force (0.86), and No weapon and Theft (0.98). A strong positive correlation signifies that when one variable increases, the other tends to increase as well.

The strong positive correlation between female victims and individuals of Black or Hispanic descent suggests targeting of these ethnic groups in crimes where females are victims. This could indicate issues related to racial, ethnic profiling, discrimination, or socio-economic factors. Furthermore, the strong positive correlation between female victims and crimes involving no weapon or theft indicates potential vulnerability of females to these types of crimes. It may also imply that females are more likely to be targeted for theft-related offenses, or that they are less likely to encounter armed assailants compared to males. Additionally, the positive correlation between crimes involving no weapon and victims of Black or Hispanic descent suggests that a

significant proportion of crimes committed without weapons involve victims from these ethnic groups. This opens the opportunity for these vulnerable groups to equip themselves with sufficient body defense to deter these crimes. Lastly, the strong positive correlation observed between violent crimes and bodily force, as well as theft crimes and incidents without weapons, underscores the necessity of implementing violence prevention strategies and interventions to reduce these crimes and ensure public safety.

3.2. EDA on Crimes in 2022

Due to the high frequency of crimes committed in 2022, this research will utilize a sample of 20,000 crimes in 2022 to analyse the possible patterns and reasons for the increased frequency of crimes. Looking at Figure 7, victims aged 20 – 40 suffer from the most crimes, with crimes occurring more frequent towards older victims than younger victims. Similarly, the proportions of crimes by sex, descent, weapons used and categories in 2022 mimic the proportions of crimes from 2020 to 2024 (Figure 8).

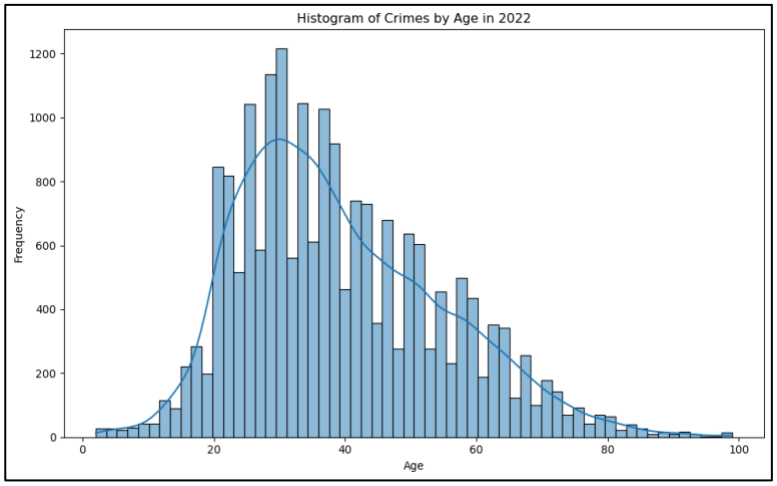


Figure 7. Frequency of Crimes by Age in Los Angeles in 2022

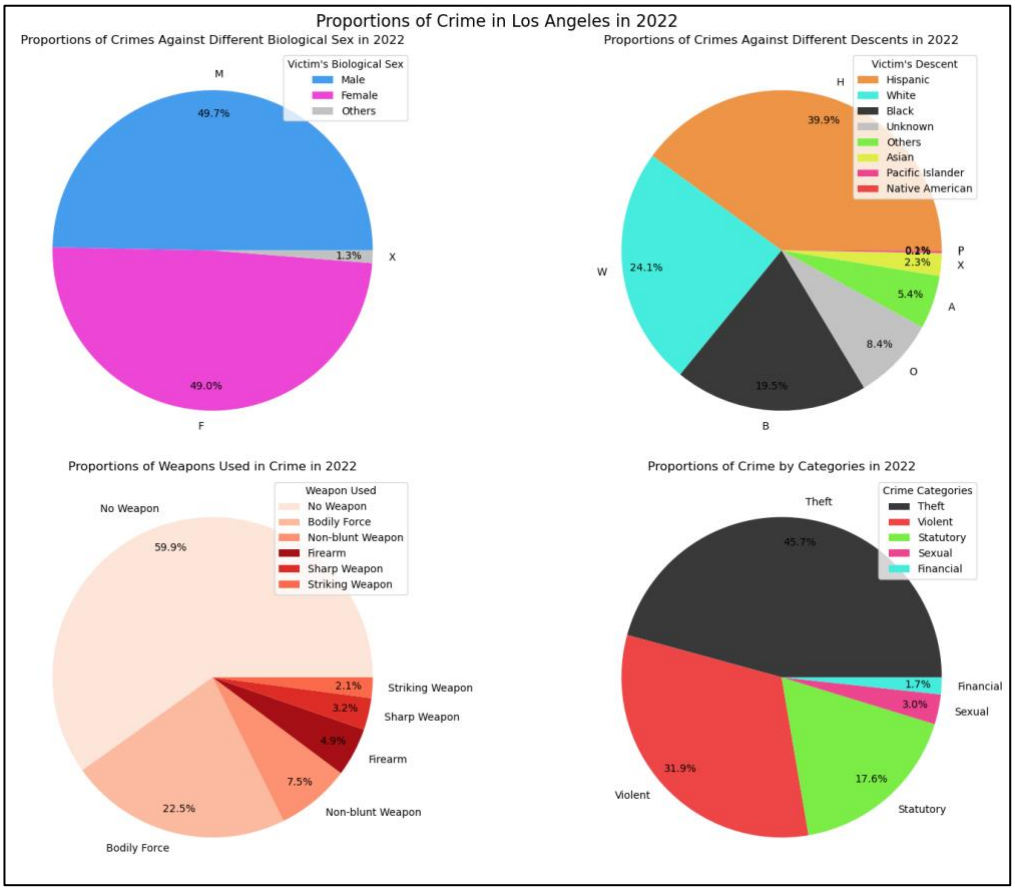


Figure 8. Proportion of Crimes by Categories in Los Angeles in 2022

Through spatial analysis of individual crimes according to their specific location, a high frequency of crimes occurs in Central LA, followed by 77th Street and Southwest as seen in Figure 9. Using the spatial analysis of crimes committed, this research hopes to predict possible locations to improve surveillance and patrols to deter crimes from happening in the vicinity.

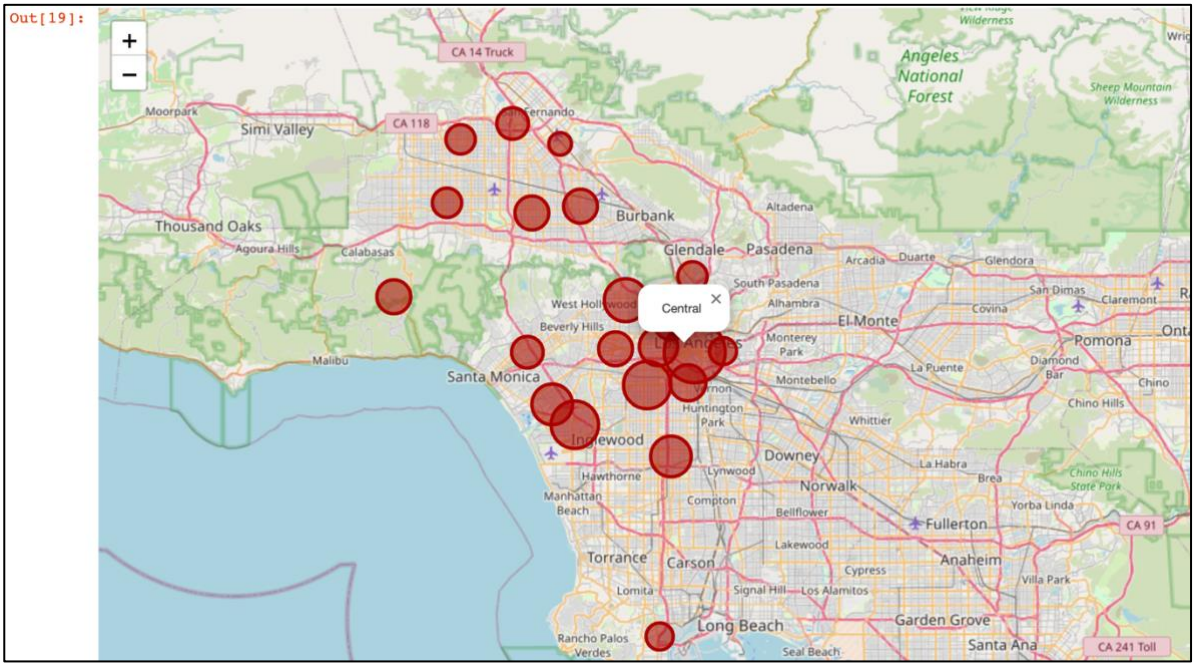


Figure 9. Frequency of Crimes by Vicinity in Los Angeles in 2022

4. Unsupervised Learning Task

Through unsupervised learning, this research hopes to find intrinsic groups of crimes committed from 2020 2024 and draw conclusions from them. To determine the optimal number of clusters that gives the best performance, the elbow method is performed (Figure 10). The elbow method describes the distortion score respective to the number of clusters. As observed, the higher the K, the lower the distortion score, where each data point is closer to their respective cluster centers (centroids). Since the distortion score decreases most significantly at K = 4 (elbow), the dataset will be split into 4 clusters (Figure 11)

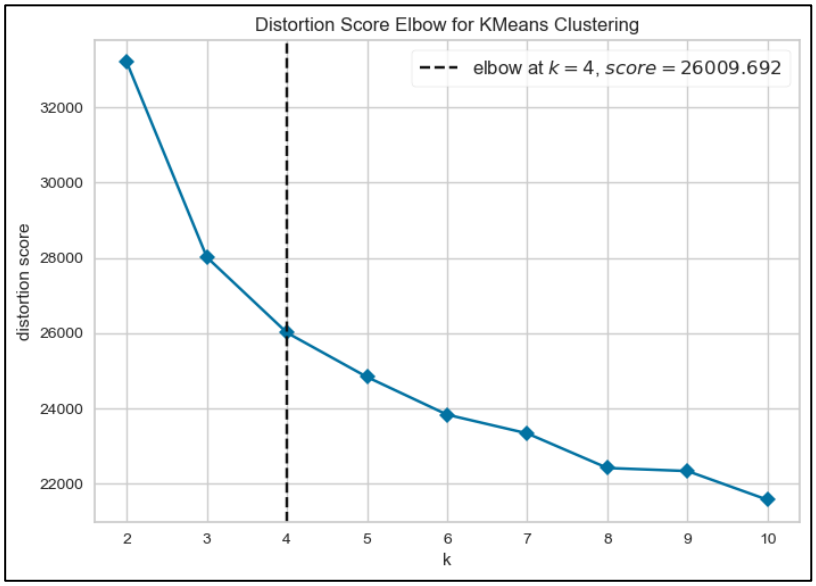


Figure 10. Elbow Method



Figure 11. K-Means Cluster

Using K-means clustering, the data points of frequency of crimes have been grouped together based on their proximity to the centroid of each cluster. By minimizing the sum of squared distances (inertia) between each data points, the model can assign data points to each cluster. From Figure 11, it can be seen that the dataset has successfully clustered into 4 centroids.

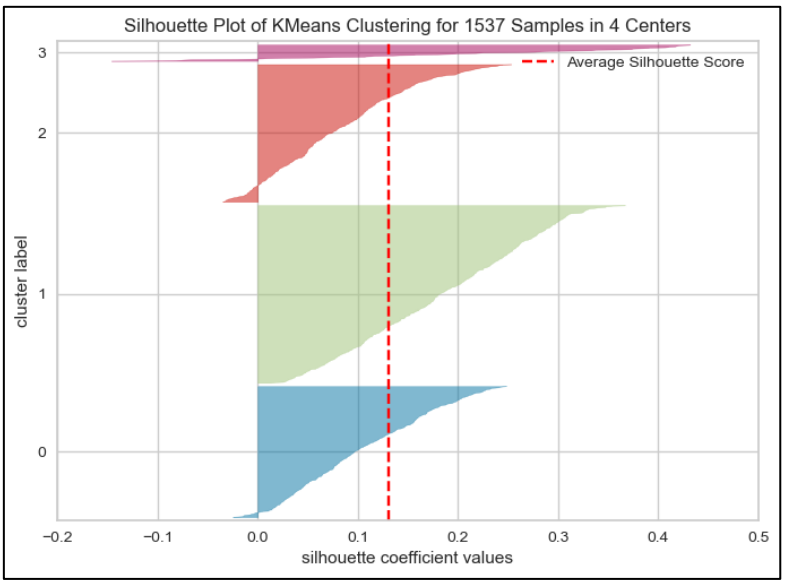


Figure 12. Silhouette Plot

However, although most data points were successfully assigned to the correct clusters, Cluster 3, 2 and 0 are occasionally incorrectly clustered. Using the Silhouette plot, we can conclude that although the model can successfully cluster datapoints accordingly, the model is not very accurate as described by the high inertia of 26,009. This means that the distance between samples to the centroids are extremely high, thus resulting in a positive but low average Silhouette score.

5. Classification Task

Through classification tasks, this research hopes to predict characteristics of victims to identify demographic groups vulnerable to crimes. The performance of different classification models will be compared to attempt predicting variables such as biological sex, ethnic descent, and age range. Based on the classification performance of different models on the prediction of biological sex, we can conclude that the logistic regression model has the highest accuracy of 62.24% (Figure 13). This means that the model was capable of successfully predicting 62.24% of instances in the test set. Since logistic regression is commonly used to predict binary outcomes with two outcomes, the model would have been more accurate if the number of outcomes were limited to two.

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
lr	Logistic Regression	0.6224	0.0000	0.6224	0.6246	0.6213	0.2587	0.2593	3.6540
et	Extra Trees Classifier	0.5972	0.0000	0.5972	0.5990	0.5965	0.2114	0.2124	1.4600
rf	Random Forest Classifier	0.5946	0.0000	0.5946	0.5970	0.5938	0.2050	0.2060	1.1890
gbc	Gradient Boosting Classifier	0.5815	0.0000	0.5815	0.5820	0.5810	0.1818	0.1822	4.6950
lightgbm	Light Gradient Boosting Machine	0.5739	0.0000	0.5739	0.5758	0.5734	0.1632	0.1635	0.7420
lda	Linear Discriminant Analysis	0.5660	0.0000	0.5660	0.5671	0.5654	0.1608	0.1610	0.3250
ridge	Ridge Classifier	0.5643	0.0000	0.5643	0.5639	0.5635	0.1450	0.1451	0.3550
knn	K Neighbors Classifier	0.5516	0.0000	0.5516	0.5527	0.5516	0.1211	0.1211	0.4830
ada	Ada Boost Classifier	0.5489	0.0000	0.5489	0.5532	0.5405	0.1160	0.1196	0.8880
dt	Decision Tree Classifier	0.5392	0.0000	0.5392	0.5404	0.5390	0.0959	0.0960	0.3160
svm	SVM - Linear Kernel	0.5058	0.0000	0.5058	0.4909	0.3672	0.0220	0.0459	1.3180

Figure 13. Comparison of Classification Models

From the Receiver Operating Characteristic (ROC) curve (Figure 14), it is observed that the AUC of class 0 (Female) and 1 (Male) are similar, however the AUC of class 3 (Others) is exceptionally high at 0.93. This means that the model is highly accurate at predicting crimes committed against 'Others'. If equal weight were given to each instant regardless of class imbalances, the model is considerably accurate with a micro-average AUC of 0.83. However, in considering the disparities in performance among the different classes, the model is less accurate with a macro-average AUC of 0.76.

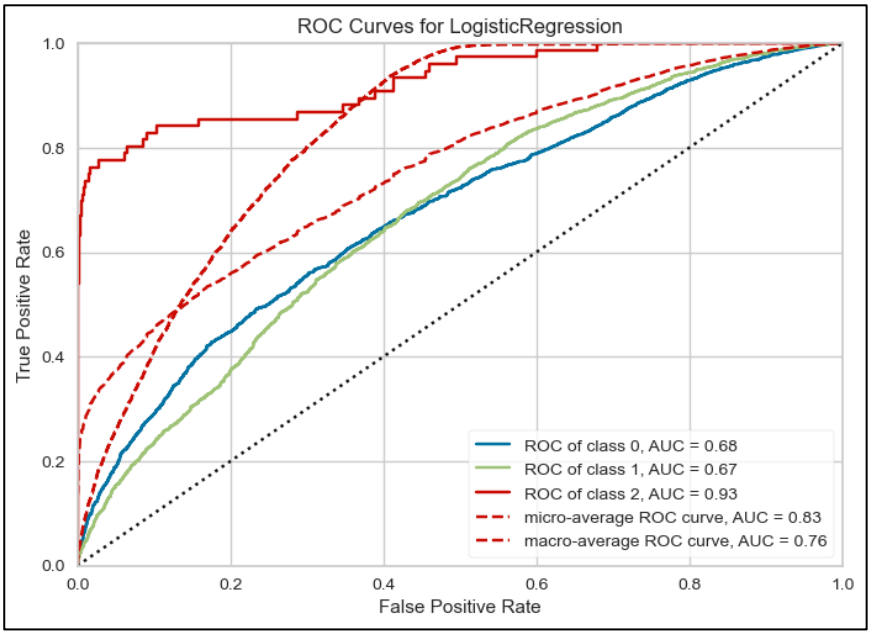


Figure 14. ROC Curves for Logistic Regression

Although attempts were made to predict victim's descent and age range, all models performed badly with accuracies of less than 0.50 and 0.30 respectively.

6. Regression Task

Through classification tasks, this research hopes to predict the latitude and longitude of crimes to identify geographic areas that are more prone to crimes. Unlike classification, the performance of regression tasks is mainly determined by the mean absolute error (MAE), coefficient of determination (R^2), and root mean squared error (RSME). Of the different models, the random forest regressor performed the best in predicting both the latitude and longitude of crimes. The random forest regressor (rf) combines the predictions of multiple decision trees to reduce overfitting of data and improve accuracy. In both latitude and longitude, the rf achieved a MAE of 0.0041 and RMSE of 0.0109 and 0.0095 respectively, showing that the average absolute mean between the predicted values and actual values is 0.0041 degrees, or an approximate residual of 644.7 meters with a small deviation. (Figure 15 & 16)

	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE	TT (Sec)
rf	Random Forest Regressor	0.0041	0.0001	0.0109	0.9895	0.0003	0.0001	4.5150
et	Extra Trees Regressor	0.0054	0.0001	0.0113	0.9891	0.0003	0.0002	4.2710
dt	Decision Tree Regressor	0.0039	0.0001	0.0117	0.9881	0.0003	0.0001	0.2940
lightgbm	Light Gradient Boosting Machine	0.0077	0.0002	0.0138	0.9835	0.0004	0.0002	0.4980
gbr	Gradient Boosting Regressor	0.0110	0.0003	0.0167	0.9760	0.0005	0.0003	1.3860
ridge	Ridge Regression	0.0133	0.0003	0.0179	0.9728	0.0005	0.0004	0.2090

Figure 15. Comparison of Regression Models (Latitude)

	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE	TT (Sec)
rf	Random Forest Regressor	0.0041	0.0001	0.0095	0.9914	0.0001	0.0000	5.2480
et	Extra Trees Regressor	0.0051	0.0001	0.0096	0.9914	0.0001	0.0000	4.3330
dt	Decision Tree Regressor	0.0043	0.0001	0.0105	0.9895	0.0001	0.0000	0.3170
lightgbm	Light Gradient Boosting Machine	0.0083	0.0002	0.0132	0.9837	0.0001	0.0001	0.3340
gbr	Gradient Boosting Regressor	0.0118	0.0003	0.0167	0.9742	0.0001	0.0001	1.4250
ridge	Ridge Regression	0.0160	0.0005	0.0213	0.9582	0.0002	0.0001	0.2930
br	Bayesian Ridge	0.0160	0.0005	0.0213	0.9582	0.0002	0.0001	0.2920
ada	AdaBoost Regressor	0.0197	0.0007	0.0256	0.9395	0.0002	0.0002	0.8490
huber	Huber Regressor	0.0428	0.0033	0.0567	0.6936	0.0005	0.0004	0.6960

Figure 16. Comparison of Regression Models (Longitude)

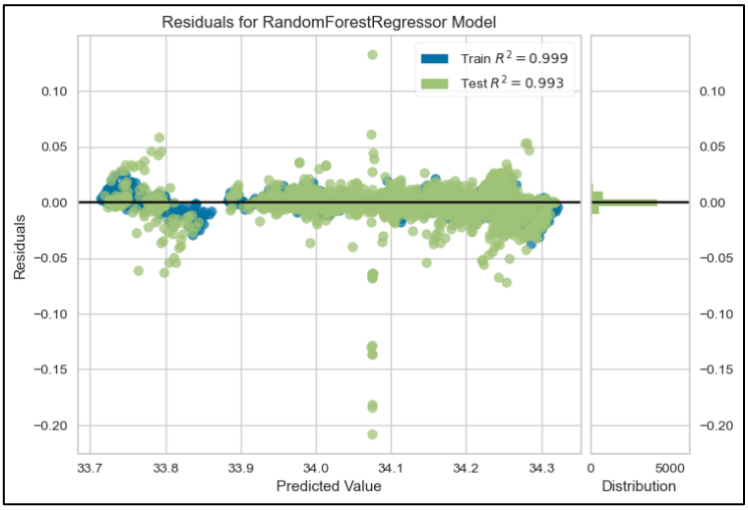


Figure 17. Residuals Plot (Latitude)

Additionally, the low R^2 of 0.9895 and 0.9914, represents the respective proportion of latitude and longitude that is explained by the features, where a value closer to 1 indicates a model that explains all variability, demonstrating a better fit and predictive power of the model. The low residual of each predicted value is further illustrated in Figure 17 and 18.

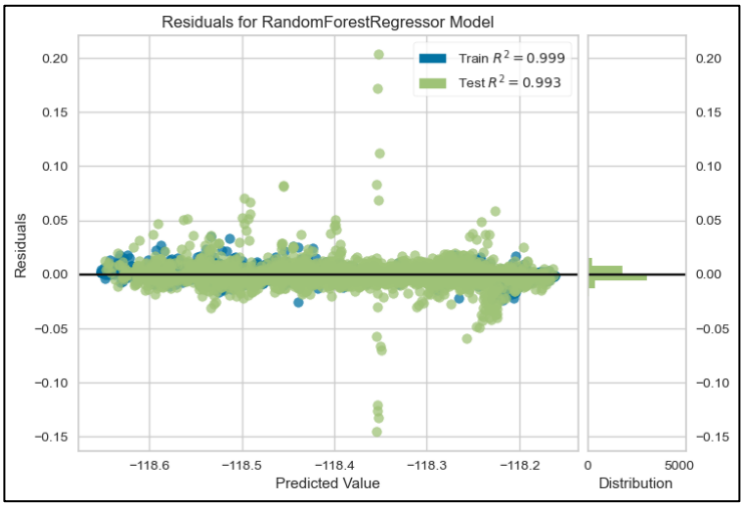


Figure 18. Residuals Plot (Latitude)

The learning curve of the models also help to illustrate a well fit model, with a respectively low training score at the beginning that gradually increases with more training examples and plateaus gradually. Moreover, the low validation score initially increases upon adding more training examples and flattens gradually, showing that adding more training examples does not improve the performance on unseen data. These observations show that the model is a good fit. (Figure 19 & 20)

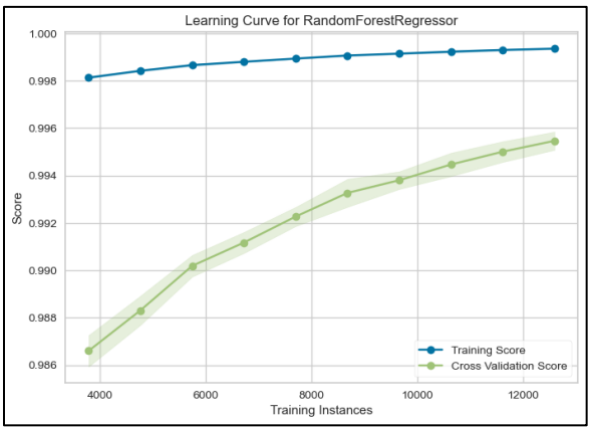


Figure 19. Learning Curve (Latitude)

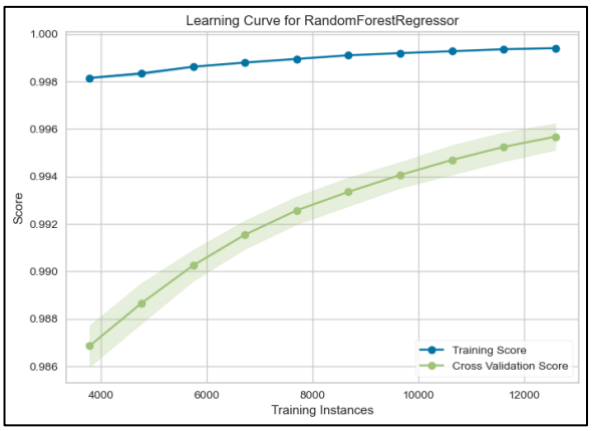


Figure 20. Learning Curve (Longitude)

7. Conclusion

In conclusion, this study utilised extensive data analysis techniques and machine learning methodologies to discover crime patterns, demographic characteristics of victims, and the nature of offenses in Los Angeles from 2020 to 2024.The findings revealed disparities in victimization rates across different demographic groups and the prevalence of specific types of crimes over the years. Notably, there was a discernible increase in the frequency of crimes against certain demographics, particularly those categorized as 'Others' in terms of biological sex and 'Unknown' in ethnic descent, shedding light on the prevalence of property crime as discussed by the US Bureau of Justice Statistics (BJS) and Federal Bureau of Investigation (FBI) (Gramlich, 2020). Moreover, machine learning techniques were leveraged to predict demographic traits of crime victims and the geographic locations of criminal incidents. While some models demonstrated promising accuracy, particularly in predicting the biological sex of victims, others exhibited limitations in predicting ethnic descent and age range, underscoring the complexity of predictive tasks. However, the regression analysis provided valuable insights for law enforcement agencies to optimize resource allocation and enhance proactive policing strategies. Ultimately, the synthesis of findings from exploratory data analysis, unsupervised learning tasks, classification, and regression analyses offers a multifaceted understanding of crime dynamics in Los Angeles. These insights not only contribute to the body of knowledge in criminology but also hold practical implications for policymakers, law enforcement agencies, and community stakeholders in devising targeted interventions to combat crime, protect vulnerable populations, and foster safer, more resilient communities .

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